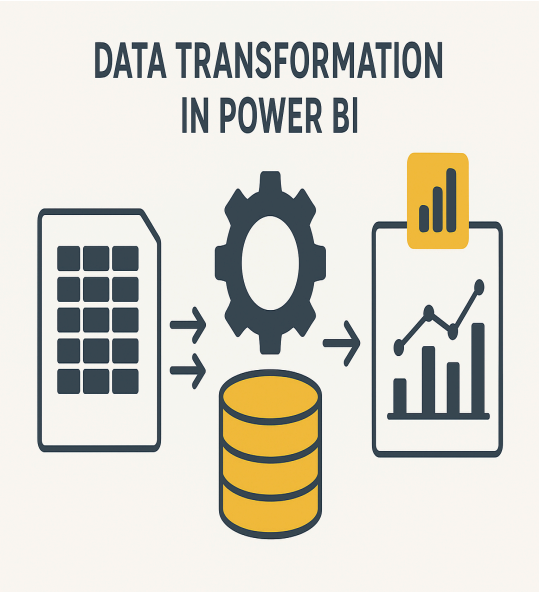
DATA TRANSFORMATION & DATA PREPARATION IN POWER BI (WITH CSV DATASET EXAMPLE)

1. Introduction to Data Transformation in Power BI

Data transformation is the foundational step in preparing raw data so that it becomes meaningful, usable, and reliable for analysis and reporting. Power BI offers an in-built tool called **Power Query Editor**, which provides a wide range of data preparation capabilities such as:

- Changing data types
- Cleaning and correcting errors
- Shaping and structuring datasets
- Merging and appending tables
- Creating new calculated columns
- Applying transformations for date, number, and text values
- Managing regional settings and formats

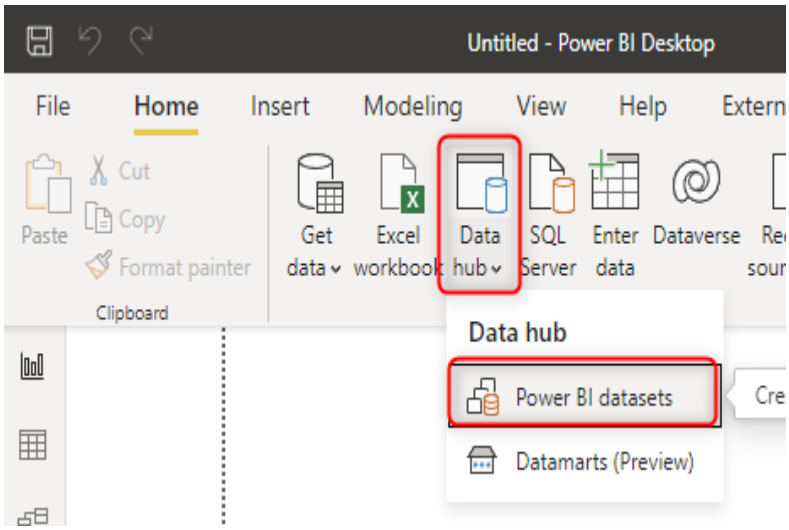
YATICORP
EMPOWERING EDUCATION



2. Connecting to a Dataset in Power BI Desktop

2.1 Opening a Blank Report

- Power BI Desktop opens into a blank workspace where you can start building reports or importing data.
- The interface may take a few moments to load, especially on first launch.



2.2 Connecting to a CSV File

YATICORP
EMPOWERING EDUCATION

Steps:

1. Click on **Get Data** on the Home Ribbon.
2. Select **Text/CSV** as the data source.
3. Browse and choose the file **clients.csv**.
4. Click **Open** to initiate the connection.

Power BI displays a **Navigator window** showing a preview of the dataset.
This helps you verify that the correct file is selected before loading or transforming it.

---

3. Overview of the Dataset

In the preview, the dataset contains the following fields:

- **client_code**
- **prefix**
- **first_name**
- **last_name**
- **date_of_birth**
- **gender**
- **email_id**
- **year_salary**
- **qualification**
- And possibly other fields depending on the dataset structure.

YATICORP
EMPOWERING EDUCATION

Power BI analyzes the sample file and automatically assigns data types such as:

- Text
- Whole number
- Decimal number
- Date
- Currency

However, sometimes the automatic detection may not be correct.
This becomes especially important when CSV data contains specific symbols or region-specific formats.

---

4. Entering Power Query Editor for Transformation

4.1 Opening the Power Query Editor

- Instead of directly loading the file into the report view, select **Transform Data**.
- Power Query Editor opens with the dataset preview on the right and the Applied Steps pane on the left.

This allows you to:

- Modify the dataset
- Apply corrections
- Clean invalid entries
- Standardize formats

YATICORP
EMPOWERING EDUCATION

4.2 Saving the Power BI File

- Use **Save As** to name the working file (e.g., *Clients.pbix*).
- Saving early helps keep your progress intact.
- You will be asked whether to apply automatically created steps — clicking **Apply** ensures synchronization.

---

5. Verifying Column Names and Column Data Types

5.1 Column Headers

Check if Power BI correctly identified the first row as header.
In this dataset, the column names are already recognized correctly.

5.2 Data Type Validation

Examples:

Column Name	Expected Type
client_code	Whole number or Text
prefix	Text
first_name	Text
last_name	Text
date_of_birth	Date
gender	Text
email_id	Text
year_salary	Currency or decimal number
qualification	Text

YATICORP
EMPOWERING EDUCATION

While most are detected correctly, two fields often require manual correction:

- **date_of_birth**
- **year_salary**

Why?
Because **regional settings** affect how dates and currency values are interpreted.

6. Importance of Regional Settings in Power BI

Power BI Desktop uses **regional settings** to interpret formats such as:

- Date
- Time
- Currency
- Number separators

Common Issue:

If the CSV file is created using **US locale**, but Power BI is configured with **India locale**, the following errors arise:

- Dollar symbol ($) becomes invalid for India-region setting.
- Date formats like MM/DD/YYYY are considered invalid.

YATICORP
EMPOWERING EDUCATION

This results in:

- Column being interpreted as **text (ABC)** instead of date or number.
- **Errors** appearing in column quality indicators.

---

7. Changing Regional Settings (When Necessary)

7.1 Scenario: Data Type Errors Appear

If the date or salary columns show **ABC type with red error values**, the issue is due to incorrect locale settings.

7.2 Steps to Change Regional Settings

1. Go to **File > Options and Settings > Options**.
2. Under **Current File**, click **Regional Settings**.
3. Change the region to **English (United States)**.
4. Click **OK**.
5. Choose **Apply Changes** if prompted.
6. Refresh the Power Query preview.

YATICORP
EMPOWERING EDUCATION

7.3 Why It Matters

- US settings correctly interpret **$** as currency.
- Date formats like **12/31/1990** match US locale.
- Ensures Power BI assigns:
    - Currency data type to salary
    - Date data type to date of birth

---

8. Using “Change Type Using Locale” to Fix Errors Manually

Even after changing regional settings globally, sometimes Power BI may not correct the errors automatically.

How to fix through locale-based type conversion:

1. Right-click the column (e.g., **year_salary**).
2. Choose **Change Type > Using Locale**.
3. In the dialog, specify:
    - Data type: *Decimal Number* or *Fixed Decimal Number*
    - Locale: *English (United States)*

This forces Power Query to interpret the data using the correct cultural settings.

YATICORP
EMPOWERING EDUCATION

Result

- Errors disappear.
- Column Quality indicator shows:
    - **Valid: 100%**
    - **Error: 0%**
    - **Empty: 0%**

This method is especially important when:

- CSV has symbols like $, £, €, etc.
- System locale differs from dataset locale.

---

9. Understanding Column Quality, Distribution, and Profile

These tools help inspect data issues visually.

Column Quality

Shows:

- Valid values %
- Error %
- Empty %

Column Distribution

Shows:

- Distinct values
- Frequency of values
- Value patterns

YATICORP
EMPOWERING EDUCATION

Column Profile

Shows:

- Statistical summary (for numbers)
- Min, Max, Count, Standard Deviation
- Data types
- Value distribution graph

10. Concatenating Multiple Columns (Merging Columns)

The dataset includes:

- prefix
- first_name
- last_name

To create a single combined name:

10.1 Steps

1. Select all three columns using **Ctrl + Click**.
2. Go to the Add **Column** tab.
3. Choose **Merge Columns**.
4. Select **Space** as the separator.
5. Name the new column **FULL_NAME**.

YATICORP
EMPOWERING EDUCATION

10.2 Result

A new column like:

Mr. John Yang
Ms. Sarah Connors
Dr. Kevin White

---

11. Text Formatting Transformations

Power BI provides several formatting options:

Transformation	Purpose
Uppercase	Converts text to ALL CAPS
Lowercase	Converts text to all lowercase
Capitalize Each Word	Converts to Proper Case (Title Case)
Trim	Removes leading and trailing spaces
Clean	Removes non-printable/hidden characters

Proper Case Example

1. Select **FULL_NAME**.
2. Under the Transform tab → **Format → Capitalize Each Word**.

YATICORP
EMPOWERING EDUCATION

This replaces the existing values (unlike Add Column which creates a new column).

---

12. Deleting Unwanted Applied Steps

Sometimes a step may create unnecessary columns or incorrect transformations.
These steps can be safely removed:

- Go to the Applied **Steps** pane.
- Click the **X** beside the step you want to undo.

---

13. Finalizing Data Transformation – Close & Apply

Once all corrections are done:

1. Go to the Home tab.
2. Click **Close & Apply**.
3. Power BI applies all transformation steps in sequence.
4. Data loads into the **Data Model** for visualization.

---

14. Summary of Key Learning Points

14.1 Importing and Previewing Data

- Use CSV import functionality.
- Understand the dataset before transformation.

YATICORP
EMPOWERING EDUCATION

14.2 Data Type Management

- Ensure data types align with the nature of the field.
- Common issues arise in date and currency columns.
- Regional settings have a direct impact on data interpretation.

14.3 Regional Settings Troubleshooting

- Incorrect locale leads to:
    - Text interpretation instead of currency/date.
    - Errors in data preview.
    - Mismatch in value parsing.

- Fix using:
    - Global regional settings
    - “Change Type Using Locale”

14.4 Creating Derived Columns

- Useful for combining or formatting data fields.
- Example: FULL_NAME created using merge columns.

14.5 Data Quality Tools

- Column quality and distribution features help detect issues.
- Helpful for early-stage data anomaly detection.

YATICORP
EMPOWERING EDUCATION

14.6 Applying Changes

- Always click **Close & Apply** to commit modifications.
- All transformations are saved as sequential steps, ensuring traceability.

15. Conclusion

This detailed walkthrough demonstrates how Power BI's Power Query Editor provides a flexible and robust data preparation environment. Key operations include:

- Importing files
- Understanding and validating auto-generated data types
- Troubleshooting timezone and regional formatting issues
- Correcting errors using locale-specific transformations
- Merging and formatting text columns
- Monitoring data quality
- Saving and applying transformation steps

By mastering these foundational Power Query skills, users can handle real-world datasets with inconsistent formats, regional variations, and raw data inconsistencies. These steps ensure clean, consistent, and analysis-ready data—forming the backbone of effective BI reporting.

YATICORP
EMPOWERING EDUCATION